

5. ANSYS Engineering Data, Mesh and Boundary Conditions

The SolidWorks geometry was transferred to ANSYS Mechanical for finite-element analysis. The structural material definition used in the supplied model is PETG-based, with Young's modulus of 2.1 GPa, Poisson's ratio 0.35 and tensile yield strength of 50 MPa.

Property	Value	Unit
Density — PETG(NylonCF)	1270	kg/m ³
Young's modulus	2.1	GPa
Poisson's ratio	0.35	—
Bulk modulus	2.333×10^9	Pa
Shear modulus	7.7778×10^8	Pa
Tensile yield strength	50	MPa

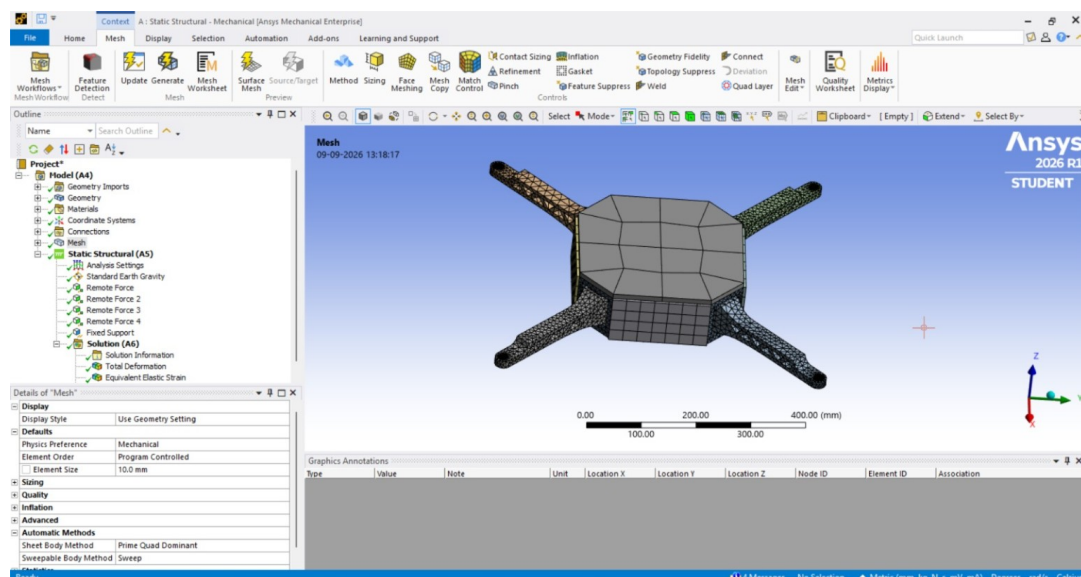


Figure 14. ANSYS finite-element mesh; element size set to 10 mm.

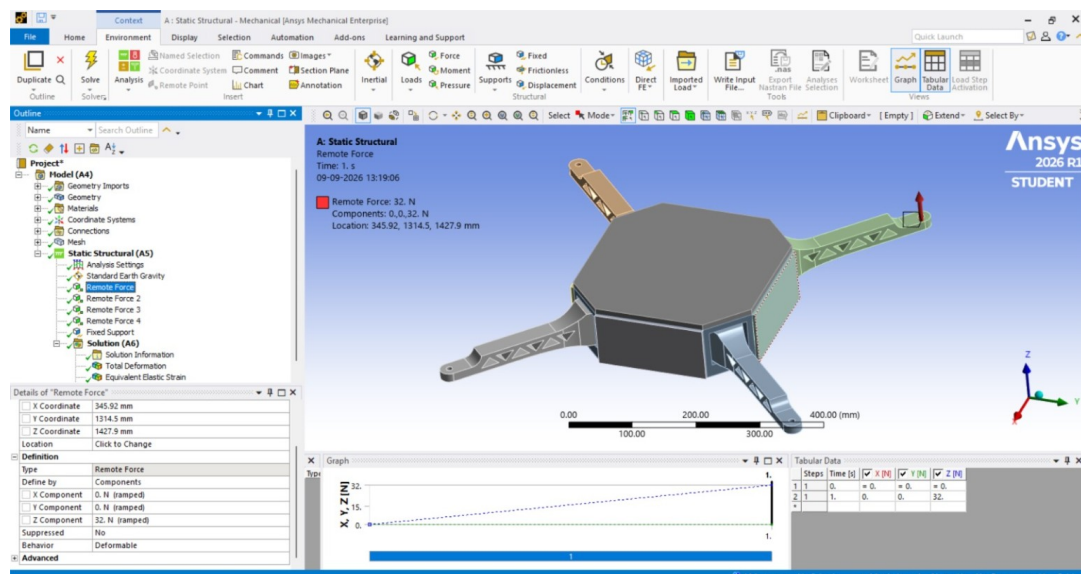


Figure 15. ANSYS structural model and loading environment.

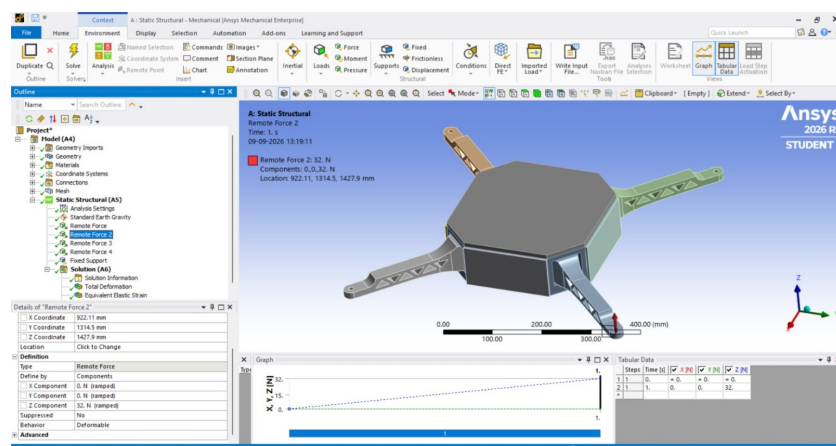
6. Structural Loading and Design Basis

Four remote forces of 32 N each are applied at the motor mounting locations in the global +Z direction. Standard Earth gravity is applied in the negative Z direction. A fixed support is applied to one face of the frame.

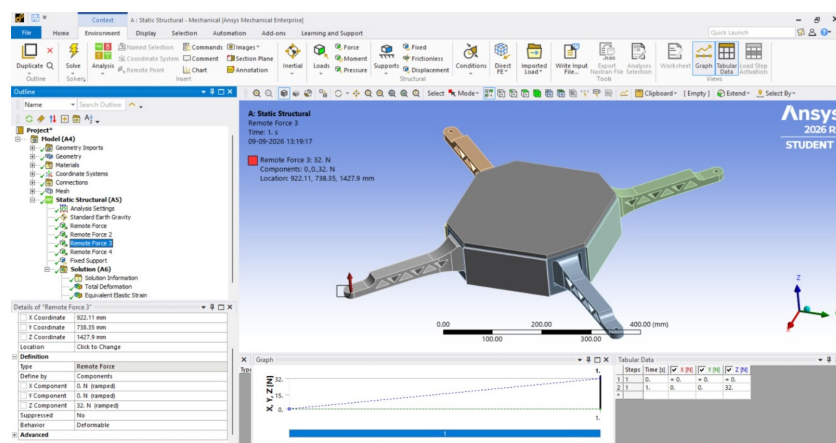
Using the SolidWorks assembly mass of 4.15 kg, the gravitational load is $W = mg = 4.15 \times 9.8066 = 40.70$ N. Equal hover load sharing would therefore require about 10.18 N per motor. The FEA does not use 10.18 N as its design load; instead, the project intentionally applies a higher 32 N per arm to provide a conservative structural load case.

The applied upward force is $4 \times 32 = 128$ N. Relative to the 40.70 N gravitational load, this corresponds to about 3.15 times the total weight, giving the model a deliberately conservative thrust-to-weight loading basis.

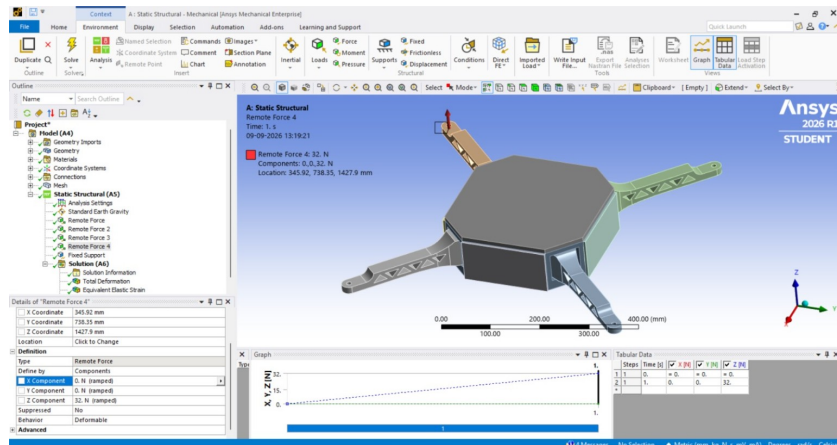
Input	Definition
Remote Force 1–4	32 N each
Direction	+Z
Total applied thrust	128 N
Gravity	−9.8066 m/s ² in Z
Fixed support	1 face
Mesh	10 mm element size



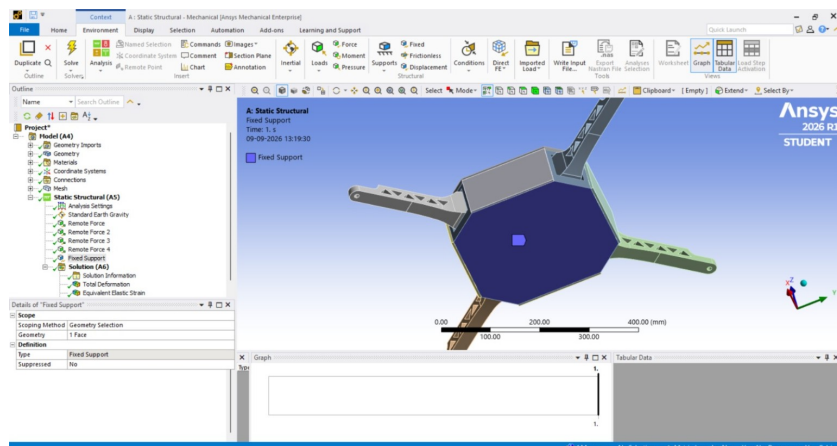
ANSYS loading/support setup — Remote Force / boundary-condition view (34661).



ANSYS loading/support setup — Remote Force / boundary-condition view (34662).



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ANSYS loading/support setup — Remote Force / boundary-condition view (34664).

7. Static Structural Results

Result	Maximum	Reference
Total deformation	0.86549 mm	0 mm at support
Equivalent elastic strain	0.00066362 mm/mm	Elastic response
Equivalent von-Mises stress	1.3927 MPa	50 MPa yield
Average von-Mises stress	0.077424 MPa	—
Factor of safety against yield	≈35.9	50 / 1.3927

The maximum von-Mises stress is only about 2.79% of the stated 50 MPa yield strength. The modeled frame therefore remains comfortably within the elastic range. The maximum deformation of 0.86549 mm occurs toward the loaded arm/motor regions, while the central body remains comparatively stiff.

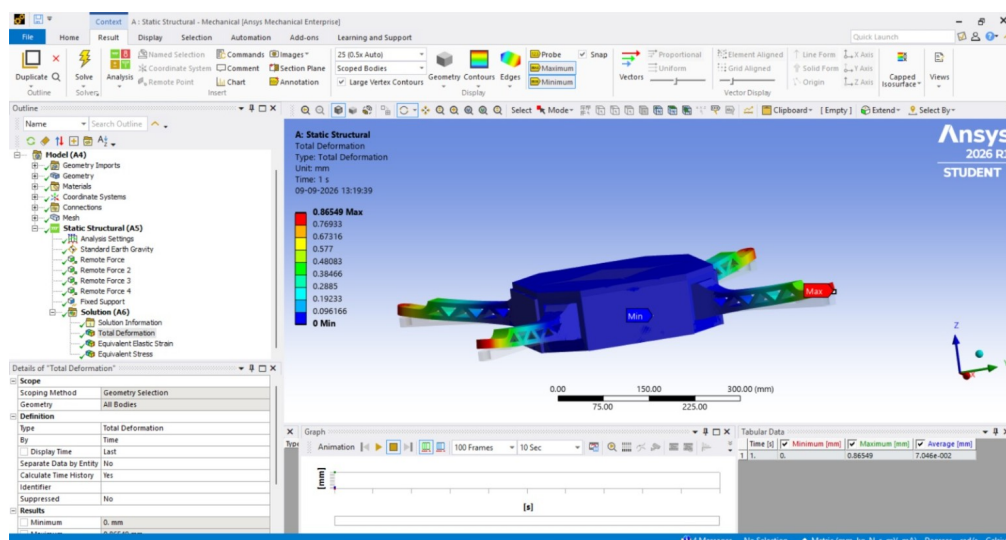


Figure. Total deformation result from the supplied ANSYS static structural solution.

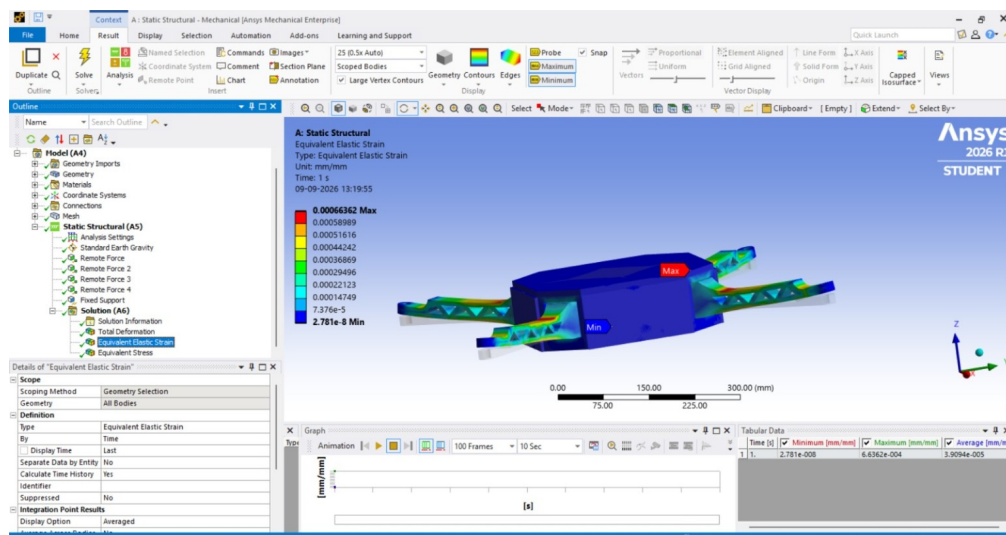


Figure. Equivalent stress result from the supplied ANSYS static structural solution.

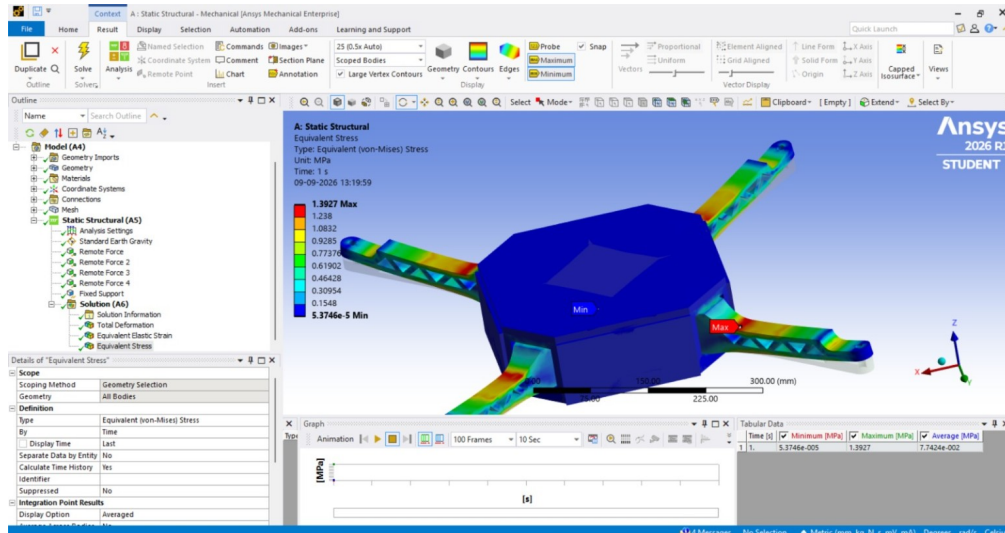


Figure. Equivalent elastic strain result from the supplied ANSYS static structural solution.

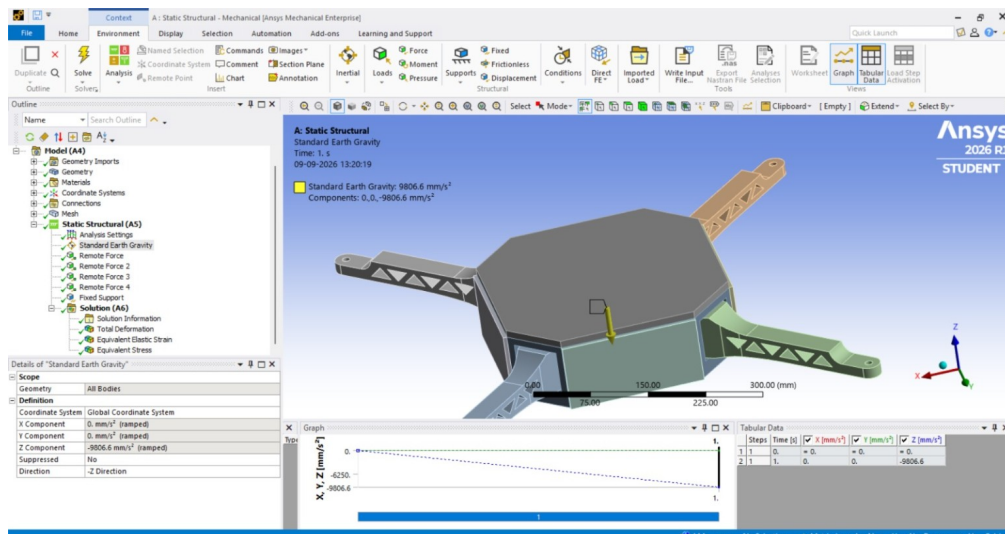


Figure. Additional static result view from the supplied ANSYS static structural solution.